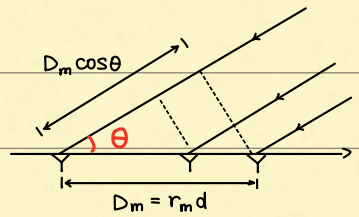
DOA M arrays, K signals



$$y_m(t) = \sum_{k=1}^{K} x_k(t)\, e^{-j\underbrace{2\pi \frac{D_m}{\lambda}}_{m_{th}\text{ array 相位差}}}$$

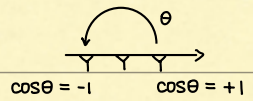
取 $d = \frac{1}{2}\lambda$

$$a(\theta)_{M\times 1} = e^{-j2\pi \frac{d}{\lambda} \vec{r}_{M\times 1} \cos\theta}$$

$0 \leqslant \theta \leqslant \pi$

Linear Array

$$A(\theta)_{M\times K} = [\, a(\theta_1) \cdots a(\theta_K) \,]$$



$$y_{M\times 1} = A_{M\times K}(\theta)\, x_{K\times 1} + e_{M\times 1} \quad (\text{SMV})$$

$$Y_{M\times L} = A_{M\times K}(\theta)\, X_{K\times L} + E_{M\times L} \quad (\text{MMV})$$

Covariance $R = APA^H + \Sigma$ $R_{M\times M} = \mathrm{Cov}(y(t)) = E[\, y(t) y^H(t) \,]$

$P_{K\times K} = \mathrm{Cov}(x(t))$, $\Sigma_{M\times M} = \mathrm{Cov}(e(t))$

Signal
- Uncorrelated $P = \mathrm{diag}(p_1 \cdots p_K)$
- Correlated : $\mathrm{rank}(P) = 1$

Uncorrelated + ULA $\Rightarrow$ APA^H Toeplitz

Pf : $R_{m_1, m_2} = \sum_{n=1}^{N} p_n e^{j2\pi \frac{d}{\lambda}(m_2 - m_1)} + \sigma_{m_1, m_2}$

only depends on $m_2 - m_1$

Noise
- WGN $\Sigma = \sigma^2 I$
- Non-Uniform Noise $\Sigma = \mathrm{diag}(\sigma_1^2 \cdots \sigma_M^2)$
- Colored Noise Σ 为一般 Toeplitz Sym

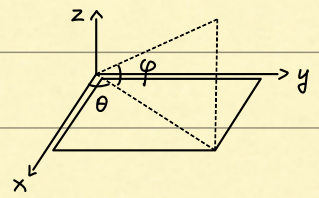
e.g. AR1 $e_{m+1} = \rho e_m + \varepsilon$,

$$\Sigma_{m_1, m_2} = \sigma^2 \rho^{|m_1 - m_2|} e^{j\phi(m_1 - m_2)}$$

Sample Cov / Empirical Cov $\hat{R} = \frac{1}{L} Y Y^H$

$L > M$, 几乎可以 $\mathrm{rank}(\hat{R}) = M$

2D-DOA

$$\underset{M^2\times 1}{a(\theta,\phi)} = e^{-j2\pi\frac{d}{\lambda}\underset{M^2\times 3}{[r_x\ r_Y\ r_z]}}\begin{bmatrix}\sin\theta\cos\phi\\ \sin\theta\sin\phi\\ \cos\phi\end{bmatrix}_{3\times 1}$$

$$= e^{-j2\pi\frac{d}{\lambda}(r_x\sin\theta\cos\phi + r_Y\sin\theta\sin\phi)}$$ 平面阵